

coefficients α , β , γ and η , but also on a random term v_t . The random term v_t reflects the inherent stochastic (probabilistic) process in an economy (for example, the unpredictable nature of human behaviour). Thus,

$$Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} + v_t \quad \dots (7.3)$$

Now, $E_{t-1}Y_t$ below is the expected or predicted value of Y made at end of time period $(t-1)$ for the future time period t based on observed values at time $(t-1)$ of variables X , Y and Z .

$$E_{t-1}Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} \quad \dots (7.4)$$

This expected value above is likely to happen or realize on average. The actual value of Y in period t , on the other hand, can be quite different depending on the random element v_t . Usually, the value of the stochastic error term is small, as it is a composite of several variables; different variables pulling Y_t away from its average value in different directions. Thus, the final value of Y_t is never totally predictable since the random term is knowable only at time t and can be thought of as the effect of human behaviour which is individual, unique and not entirely predictable.

The random error term $v_t = (Y_t - E_{t-1}Y_t)$ is also the expectational error of the rational expectations prediction. We can say that the distribution of v_t has mean zero and known variance σ^2 . Thus, on average, the expectational error will be zero (because if not, we can change the value of α to make it so). The rational expectations value will also be the most accurate expectations since it is based on the true model on which the concerned variable is based.

Let us consider an example: a simple model of consumption may have C_t depending on past wealth, past consumption and past disposable income so the rational expectations value of consumption will be the average or expected consumption level of a set of people. However, each individual's future consumption level will not and cannot be exactly predictable – there will be some variation around the average or expected consumption due to different preferences, mostly close to the average.

According to the rational expectations hypothesis, economic agents in the model (workers, consumers, firms, the Government) are knowledgeable about the true model. They have learnt through observation the behaviour of all the relevant variables (X , Y , and Z) and parameters (α , β , γ and η) in explaining Y . They know the functional form (linear here; could be quadratic, exponential or some other type) of the relationship. Thus, an individual with rational expectations can predict everything predictable about the future values of Y . The only expectational error to emerge then will be due to the inherently unpredictable portion of the variable.

Rational expectations hypothesis can be criticized for its assumption of rationality and availability of full information. People, in general, are not aware of the economic processes that take place in an economy. To assume that people can figure out the explanatory variables of the concerned variable and that they

have all relevant information is unrealistic. However, people these days have access to readily available forecasts by experts via multiple public media. Certain relevant information may be expensive to procure. The rational expectations prediction will still be correct on the average, though it may not be as accurate as it could have been with the variable.

Despite these criticisms, to use all relevant available information to form rational expectations when making decisions is nevertheless an appealing underlying microfoundation for macroeconomic theory. It assumes that individual agents optimize and make decisions that are rational and most beneficial to them. It also leads to the outcome that with perfect information and rational expectations there is market-clearing. This is due to the fact that each agent (firms and workers) is making decisions. Firms decide on the quantity of output to produce on the basis of a production function while workers decide on the amount of labour to supply on the basis of a utility function. Notice that microfoundation is an integral part of rational expectations hypothesis. With flexible prices and wages, demand and supply for output and labour will match each other in their respective markets. Thus, there will be no involuntary unemployment or no deviation of output from the natural level of output.

We need to see whether the outcomes as predicted by rational expectations at the micro level can explain macroeconomic relationships, e.g., between unemployment and inflation. One key implication of rational expectations is that government policy to reduce unemployment (monetary expansion or monetarization of debt due to fiscal expansion) will not have the desired effect if it is predictable or announced in advance. Monetary expansion to increase aggregate demand and raise output, will not lead to higher employment if workers with rational expectations anticipate that all prices (nominal wages and prices of what they buy) will increase proportionately and that their real wages are not going to increase.

A question arises at this point: How would rational expectations explain observed booms and recessions? During a boom (recession) unemployment rate decreases (increases), while actual output deviates from natural output. We will discuss these issues in the next sections while discussing the 'Lucas supply function' and the 'policy ineffectiveness proposition'.

7.6 LUCAS SUPPLY FUNCTION

According to Lucas, economic agents care about their relative price, i.e., about the price that they will receive for what they sell, relative to the price that they will pay for what they buy. Thus a firm cares about the price of the output it is selling, relative to the price of inputs that it buys, while a worker cares about the real wage rate, i.e., the nominal wage, which is the price for the labour services that he/she sells versus the prices of the goods that he/she will buy. An increase in the firm's relative price will then bring forth an increase in the output of firms while an increase in the real wage will elicit higher labour supply (assuming the substitution effect is greater than the income effect in the labour supply decision).

However, Lucas also postulates that agents are unlikely to have perfect information on their relative prices because they know much more about the price of what they are selling than about the goods or services they will buy since there are several inputs/ goods that they will buy relative to what they specialize in and sell. Thus, firms and households have local information and not global information. This can 'fool' them into making incorrect decisions (from their welfare point of view). Suppose, there is an increase in the price of the good they sell (or labour service), then each agent (firm/ worker) needs to determine whether the increase is only or mainly in the price of their good/ labour or there is a universal increase in prices of all the goods. Each agent thus has a 'signal extraction problem' to distinguish between a relative prices increase vs. an increase in the aggregate price level.

Lucas further postulates that economic agents face a second signal extraction problem. Temporary increases in relative prices are more important than permanent increases. Firms and workers need to be quick in taking advantage of profitable opportunities. If in the past, most increases in selling prices have been partly increases due to general inflation and partly due to relative price increase, then both firms and workers observing a general price increase will assume that some of the price increase is a relative price increase for their product/ labour service and will increase their supply of it.

Now, if there is general increase in the price level, say due to an increase in the money supply, and all agents knew this, with rational expectations, they would understand that all prices had increased proportionately and there is no relative prices change, then there would be no change in labour supply or in output. Thus any anticipated, expected or announced change in the aggregate price level will not lead to any change in output. However, if there is a 'price surprise', i.e. a difference between actual price level and the expected price level, then output (or labour supply for workers) will increase.

We can thus write the Lucas Supply Curve as:

$$Y_t/Y_p = f(P_t/E_{t-1}P_t) \quad \dots (7.5)$$

where Y_t = output in period t ;

Y_p = potential output (i.e., full employment output);

P_t = price of product; and

$E_{t-1}P_t$ = price of product in period t expected one period prior, i.e., in period $(t-1)$.

In logarithmic terms, we can write equation (7.5) as

$$y_t - y_p = f(p_t - E_{t-1}p_t) \quad \dots (7.6)$$

where the variables in lower case indicate the variables in logarithmic term.

Equation (7.6) indicates that output is higher than the full employment (or natural level of) output when there is a positive price surprise reflecting a relative price increase for the product (or labour service) being sold.

In an environment where prices have been stable, and most price increases have meant a relative price increase for the product/ service being sold by the firm/worker, both firms and workers can be ‘fooled’ into supplying more output/labour supply if the money supply is increased suddenly. On the other hand, if the prices have generally been volatile (as in some Latin American economies with high inflation), monetary expansion will most likely have no impact on output or labour supply since agents will not be fooled into thinking that a price increase is for their product alone, but will realize that the price increase is for all products due to general inflation.

If there was perfect information so that people could distinguish between general (i.e., aggregate) price increase vs. a relative price increase. In other words, if expected price increase mirrored the actual price increase, there would be no increase in output for firms or increase in labour supply for workers. But since both firms and workers get ‘fooled’ into thinking that *their* price has increased (because they only have local information on their sale price and not global information on all prices), that all firms and workers tend to increase output and labour supply. According to Lucas, such ‘fooling’ behaviour leads to fluctuations in the level of output. This explains business cycle in an economy. Note that unlike the AS-AD approach of Section 7.2 where firms knew the correct real wages but workers did not, in the Lucas Supply Curve both the agents (firms and workers) are symmetrically ‘fooled’,

The Lucas supply function relies on a price surprise to explain why output fluctuates from the ‘full employment’ or ‘natural level of output’ level. Had there been perfect information, so that prices could be anticipated accurately, agents would be able to distinguish between increases in nominal vs. real price increases and we would get the results of the classical economists i.e., dichotomy between the real and monetary sectors of the economy and that money is neutral.

7.7 POLICY INEFFECTIVENESS THEOREM

While the Lucas supply function focused on the supply side, the policy ineffectiveness theorem focuses on the demand side to explain deviations of output from the natural level of output. Here again, these deviations of output and employment are explained by price surprises caused by the non-anticipatable portion of money change. The model here is based on the seminal paper “Rational expectations and the theory of economic policy” by Sargent and Wallace (1976) [see, Sheffrin (1996)]. Note that only monetary and no fiscal changes are being considered here.

Let us assume that economic agents have rational expectations. Therefore, they have learnt from past observation how the monetary authorities operate, i.e., what is their money supply rule (see Unit 18 for further discussion on this). However, there is a random, unpredictable portion to the money supply since it is also dependent on factors beyond the central bank’s control (e.g., capital flows in and out of the country; banks may keep reserves higher than the required reserve ratio; public may decide to hold more cash than bank deposits). According to the

'policy ineffectiveness theorem', output will not deviate (from the full employment level) with a change in money supply unless it is unpredictable. Any predictable change in money supply will lead to a change in expected prices such that there will be a corresponding change in actual prices, but no change in output. Prices, on the other hand, will be influenced by both predictable and unpredictable parts of any money supply change. The model consists of an aggregate demand relation, aggregate supply relation and a money supply rule, along with an equation capturing rational expectations:

1) Aggregate Demand Relation:

$$M_t + V_t = P_t + y_t \quad \dots (7.7)$$

Equation (7.7) is the familiar $MV=PY$ equation of classical theory except that each term above is in log form and therefore is a linear equation.

M_t = log of the money supply

V_t = log of the (assumed constant) velocity of money

P_t = log of the price level

y_t = log of real output

Allowing V_t to vary with interest rate is not critical to the argument so the assumption that it is constant is for simplification.

This can be seen as the AD line in Fig. 7.2. An increase in M_t will shift the AD line to the right.

2) Aggregate Supply Relation:

$$y_t = y_p + \beta(P_t - {}_{t-1}P^e_t) \quad \dots (7.8)$$

Where

y_p = log of full employment output

${}_{t-1}P^e_t$ = log of the price level expected in period t , with the expectation formed in period $(t-1)$

This is based on the Lucas Supply function. Output will differ from the full employment level of output if there is a price surprise, i.e., the price level in period t was not anticipated. In Fig. 7.2, this is the AS line. It shows that $y_t = y_p$ if $P_t = {}_{t-1}P^e_t$. However, y_t increases relative to the fixed y_p if P_t is larger than the expected ${}_{t-1}P^e_t$ (that is, when there is a positive price surprise).

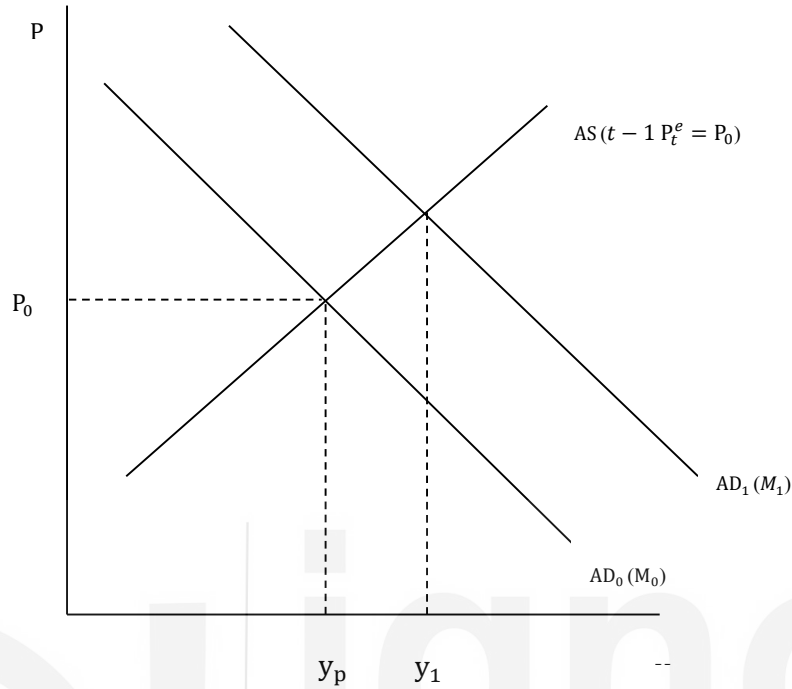


Fig. 7.2: Lucas AS Curve

3) Money Supply Rule:

$$M_t = \alpha y_{t-1} + \epsilon_t \quad \dots (7.9)$$

where $E(\epsilon_t | I_{t-1}) = 0$

Implication of equation (7.9) is as follows: The monetary rule has a portion αy_{t-1} that is under the control of the monetary authorities. This portion is based on past observed data which is known to the general public as well and can be anticipated by them. However, there is a random component, ϵ_t in equation (7.9) which nobody can know before the end of period t and cannot be predicted. All we can say is that the expected value of this random component is zero given the information $I_{(t-1)}$ (available at the end of period $t-1$) on the basis of which any prediction for period t can be made.

4) Rational Expectations:

$${}_{t-1}P_t^e = E(P_t | I_{t-1}) \quad \dots (7.10)$$

Equation (7.10) captures the following: With rational expectations, the expected value of price in period t (which is based on information I available in period $(t-1)$) is endogenous. Endogenous means that the variable (in this case, prices) is derived within the model. Thus, the same model that explains prices is used to estimate expected prices. The price level in period t that people 'feel' will prevail or psychologically expect (that is, ${}_{t-1}P_t^e$) is obtained from the model as the mathematical expectation

of prices in period t based on the information available in period $(t-1)$, that is, $E(P_t|I_{t-1})$.

Before we move to the mathematical derivation of the policy ineffectiveness theorem, let us graphically see its results. In Fig. 7.2, if money supply is increased (based on the money supply rule given at equation (7.9)), the AD curve $AD_0 (P_0 = M_0 + V - y_p)$ will shift to $AD_1 (P_t = M_t + V - y_t)$. In the absence of any change in expected future prices, the AS curve (drawn for a particular expected price) will remain unchanged, and the shift in the AD curve will result in increased output as well as prices, from y_p to y_1 . This is consistent with the original Phillip's Curve. It shows a trade-off between lower unemployment (increased output) and price rise.

Given rational expectations, in period $(t-1)$ itself, people will anticipate the shift of the AD curve due to the anticipated or expected component (not the random portion) of the money supply rule. Therefore, the AS curve will shift up corresponding to the price increase expected due to this expected component. Such a shift in the AS curve is shown in Fig. 7.3 with the dotted AD curve (which is only the anticipated and not the actual increase in the price level).

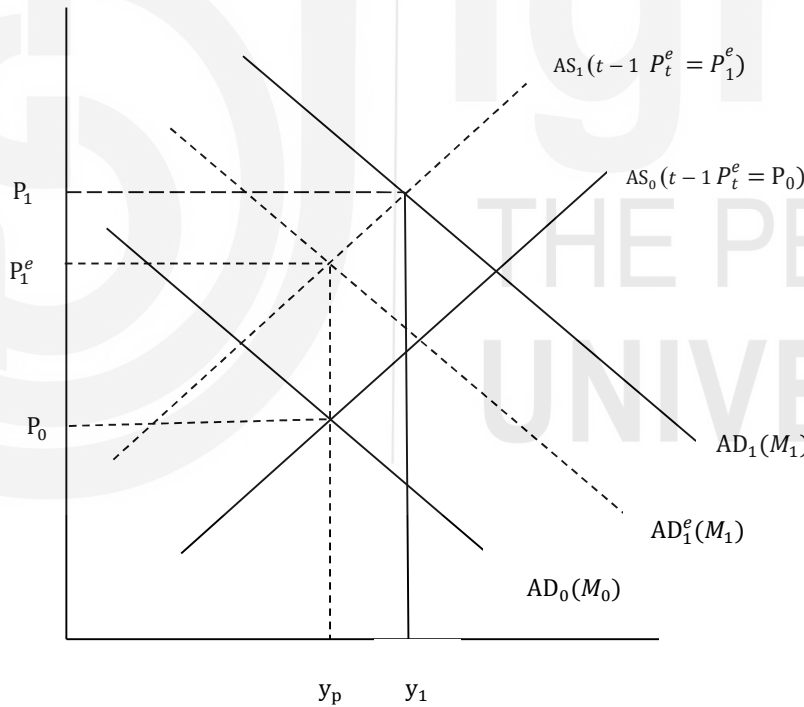


Fig. 7.3: AS-AD Model with Rational Expectations

If the random term ϵ_t in the money supply rule were zero, the actual price level in period t would be exactly as anticipated; there would be no price surprise in the Lucas supply function. Therefore, there is no change in the output level. However, actual money supply M_t can be different from the expected money supply M_t^e due to the random term ϵ_t . In Fig. 7.3, we have shown the difference to be positive, as a result of which AD_1 is to the right of the dotted AD_1^e . The

actual price level can be different from the expected price level ${}_{t-1}P_t^e$ and due to this price surprise, the actual output will differ from the full employment level of output y_t .

Thus, both the anticipated and unanticipated portions of the increase in money supply result in an increase in the price level. However, only the unanticipated portion of the money supply increase which shifts the AD curve unexpectedly leads to the price surprise, and therefore to a change in the output level.

Now, let us combine the four equations (7.7), (7.8), (7.9) and (7.10) together.

$$1) \text{ Aggregate Demand Relation: } M_t + V = P_t + y_t \quad \dots(7.7)$$

$$2) \text{ The Aggregate Supply Relation: } y_t = y_p + \beta(P_t - {}_{t-1}P_t^e) \quad \dots(7.8)$$

$$3) \text{ The money supply rule: } M_t = \alpha y_{t-1} + \epsilon_t \quad \dots(7.9)$$

$$4) \text{ Rational Expectations: } {}_{t-1}P_t^e = E(P_t | I_{t-1}) \quad \dots(7.10)$$

Let us substitute y_t from (7.8) and M_t from (7.9) in (7.7). We get

$$\alpha y_{t-1} + \epsilon_t + V = P_t + y_p + \beta(P_t - {}_{t-1}P_t^e) \quad \dots (7.11)$$

Now let us take the mathematical expectation of the above equation as of time (t-1) to get ${}_{t-1}P_t^e$

$$\alpha y_{t-1} + E(\epsilon_t) + V = {}_{t-1}P_t^e + y_p + \beta({}_{t-1}P_t^e - {}_{t-1}P_t^e).$$

This simplifies to

$$\alpha y_{t-1} + V = {}_{t-1}P_t^e + y_p \text{ or that}$$

$${}_{t-1}P_t^e = \alpha y_{t-1} + V - y_p \quad \dots (7.12)$$

From (7.7) we know that $P_t = M_t + V - y_t$. Substituting for M_t from (7.9), we get

$$P_t = \alpha y_{t-1} + \epsilon_t + V - y_p + \beta(P_t - {}_{t-1}P_t^e) \quad \dots (7.13)$$

Subtracting ${}_{t-1}P_t^e$ in equation (7.12) from P_t in equation (7.13), we obtain:

$$P_t - {}_{t-1}P_t^e = \epsilon_t + \beta(P_t - {}_{t-1}P_t^e) \quad \dots (7.14)$$

Equation (7.14) can be re-arranged to show that the gap between the actual and expected price level is

$$P_t - {}_{t-1}P_t^e = \epsilon_t / (1 + \beta) \quad \dots (7.15)$$

From the Lucas Supply Function given at equation (7.8) we find that

$$y_t = y_p + \beta(P_t - {}_{t-1}P_t^e)$$

which means

$$y_t = y_p + \beta \epsilon_t / (1 + \beta) \quad \dots (7.16)$$

Equation (7.16) implies that output (y_t) is higher than full employment output (y_p) only if there is some unexpected or unanticipated increase in the price level. From (7.15), we find that $P_t = {}_{t-1}P_t^e + \epsilon_t / (1 + \beta)$. It means that both the expected price level and the unanticipated portion of the money supply affect the price

level. These are the two results we set out to show in the policy ineffectiveness proposition.

Only unanticipated (or, surprise) changes in monetary policy can influence output, given rational expectations and thereby knowledge of how the monetary authorities react to a given situation through the money supply rule. This money supply rule could have taken a different functional form, but to the extent that it is known, it cannot have any effect since the public anticipates it and changes expectations accordingly. The policy ineffectiveness proposition thus negates the possibility of any systematic policy rule to keep the actual output level of the economy above its potential output level.

Both the Lucas Supply Curve and the Policy Ineffectiveness Proposition indicate that money is not neutral in the short run. This is due to the fact that unanticipated high prices fool the economic agents to voluntarily work harder or increase output. Keynesian economists criticize this approach by arguing that correct information should be available within a short time, so that booms and recessions spanning several years are not well explained by rational expectations. They also criticize the assumption that all markets clear (which implies that people are voluntarily unemployed) when there is a high rate of unemployment.

Despite the criticism, we can say that rational expectations is an appealing way to incorporate microeconomic foundations of macroeconomic relations and variables. This has several interesting implications. In consumption, for example, the random walk of consumption and the issue of Ricardian Equivalence.

Check Your Progress 2

- 1) Bring out the salient features of the Rational Expectation Hypothesis. Explain it by considering three variables.

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- 2) Describe the Lucas supply curve.

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- 3) What is meant by policy ineffectiveness theorem?

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7.8 LET US SUM UP

In this unit we learnt about the relation between price and unemployment, the short-run and long-run impact of money supply on output and prices by discussing the non-neutrality of money, traditional and modern Phillips curve, and the adaptive expectations. We also learnt about the Lucas critique and the perfect foresight as related to the effect of monetary policy in an economy. In the perfect foresight model, we learnt that the prediction is accurate by which the expected error is zero. However, an economic variable may be dependent on a random component that reflects the inherent stochastic process of the variable, on which the rational expectations hypothesis is based on. Then we discussed the formation of rational expectations hypothesis, its properties and implications. We also learnt about the Lucas Supply function, which focused on the supply side. In contrast, the policy ineffectiveness proposition focuses on the demand side to explain deviations of output from the natural level of output that are explained by price surprises caused by the non-anticipatable portion of money change.

7.9 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) The AD function: $Y = f(G, T, M/P)$ is an increasing function of G , T and M/P , where G = Government expenditure, T = Taxes; M/P = nominal money/price level. The AS function is $P = P^e (1 + m)f(u, z)$. Use Fig. 7.1. Refer to Section 7.2.
- 2) The traditional Phillip's Curve shows the trade off between price rise and unemployment, which is based on the same phenomenon of assuming that price expectations of workers are unchanged even when a price rise is observed. The 'modified' Phillip's Curve shows that price expectations rise when price is greater than expected price, so only a price rise higher than expected will lead to an increase in employment or output, Y . Use Fig. 7.1.
- 3) Refer to Section 7.3 and answer.
- 4) There is the 'perfect foresight' if the true AS-AD model equations and parameters are known. It implies that people have perfect information on the true model that generates actual P . If the same model is used to get the value of P^e (expected price), then P^e should equal P . That means there will be no expectational error (that is, $P - P^e = 0$).